LELEC2870

Intermediate report - Linear model

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Team 6
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1. INTRODUCTION

The purpose of this report is to detail our approach to using a linear model to predict cardiac risk in Smurfs. We will review the preprocessing performed based on our EDA, the model chosen, the influence of hyperparameters on the results of our best model, and we will conclude with a summary.

2. PREPROCESSING

Our preprocessing pipeline was guided by exploratory data analysis (EDA) and comprises six main steps applied in the following order:

(1) Removal of irrelevant features: The `img_filename` column has been removed because images will be processed separately in Part 3.

(2) Outlier management: We observed very few outliers in the dataset as a whole, but we decided to cap blood pressure values at 150 mmHg after observing the distribution of this feature.

(3) Feature engineering: We created a new BMI feature calculated as BMI = $\frac{\text{weight}}{(\text{height}/100)^2}$. This transformation captures the nonlinear relationship between weight and height while reducing the observed multicollinearity (height-weight correlation of 0.33). The original height and weight features are retained in this first part.

(4) Encoding categorical variables: Ordinal features (sarsaparilla, smurfberry liquor, smurfin donuts) were encoded using OrdinalEncoder (Very low = 0 to Very high = 4) but via manual mapping. The profession feature was encoded as one-hot with the first category removed (`drop_first=True`) to avoid multicollinearity, as EDA showed that this variable had a negligible impact on cardiac risk (very similar averages between occupations: 0.07-0.08).

(5) Scaling: We applied a scaler on our numerical features. In accordance with best practices, the `StandardScaler` was only fitted on the training set and then applied to the validation and test sets to avoid any data leakage.

3. FEATURE SELECTION

No feature selection was applied for this first part. Following preprocessing, we have 18 features (13 original numerical features + BMI + 3 encoded ordinal features + 5 dummy variables for occupation after one-hot encoding). Two observations from the EDA led us to make this decision:

- Most features show weak to moderate correlations with the target ($|r| < 0.052$), suggesting that they all provide potentially useful information;

- The VIF (Variance Inflation Factor) analysis did not reveal any critical multicollinearity after adding BMI and one-hot encoding.

4. MODEL SELECTION

Unlike Lasso, the **Ridge** model retains all features with reduced weights rather than eliminating them, which is preferable given the low individual correlations observed. In addition, two factors led us to select it: (1) EDA revealed mostly linear relationships between the features and the target, (2) Ridge's L2 regularization effectively manages residual multicollinearity between certain features (cholesterol-blood pressure: r=0.48).

The hyperparameter α was optimized using 5-fold GridSearchCV (see Figure below). The best alpha value found was $\alpha = 0.1$, which offers a cross-validation RMSE of 0.057769.

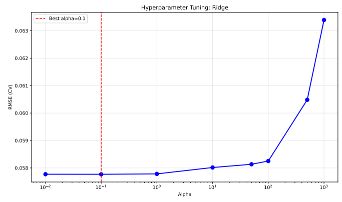

Figure 4.1: Hyperparameter-tuning

Table 4.1: Comparison of linear model performance (RMSE)

Model	Train RMSE	Validation RMSE	CV RMSE
Linear	0.0557	0.0499	0.0578
Ridge	0.0557	0.0498	0.0578
Lasso	0.0831	0.0710	0.0831

5. RESULTS

The final Ridge model, optimized with $\alpha = 0.1$, achieved an **RMSE of 0.0557 on the training set**, **0.0498 on the validation set**. The small difference between training and validation errors confirms that the model generalizes well and does not overfit.

As shown in Figure 4.1, the RMSE curve stabilizes for α values below 1, indicating that stronger regularization does not improve performance. This suggests that the dataset's relationships are predominantly linear, and the L2 penalty mainly serves to smooth the coefficients rather than to increase accuracy.

The analysis of the Ridge model coefficients (Figure 5.1) reveals a clear hierarchy of feature importance. The three most influential biomarkers are: **(1) BMI** , **(2) vitamin D** and **(3) height**.

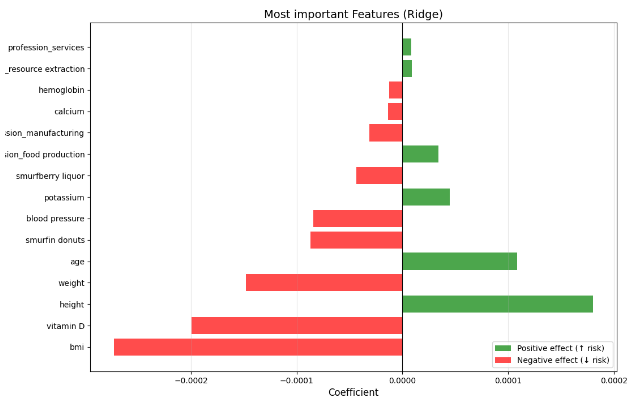

Figure 5.1: Feature Importance

6. CONCLUSION

This work presented the implementation and evaluation of a regression model to predict cardiac risk in Smurfs. Building on preprocessing and feature engineering steps guided by EDA, the linear Ridge model provided stable results comparable to Linear Regressor model.

The comparable performance between different linear approaches confirms that the underlying relationships in the data are a significant linear but exhibit weak correlations with the target variable. Therefore, we anticipate that **non-linear models (Part 2)** and future parts will be essential to capture more complex patterns and interactions within the data, potentially leading to significant performance improvements.